THE CHARIOTEER

Auriga


SIZE RANKING 21

BRIGHTEST STAR Capella (α) 0.1

GENITIVE Aurigae

ABBREVIATION Aur

HIGHEST IN SKY AT 10 P.M. December–February

FULLY VISIBLE 90°N–34°S

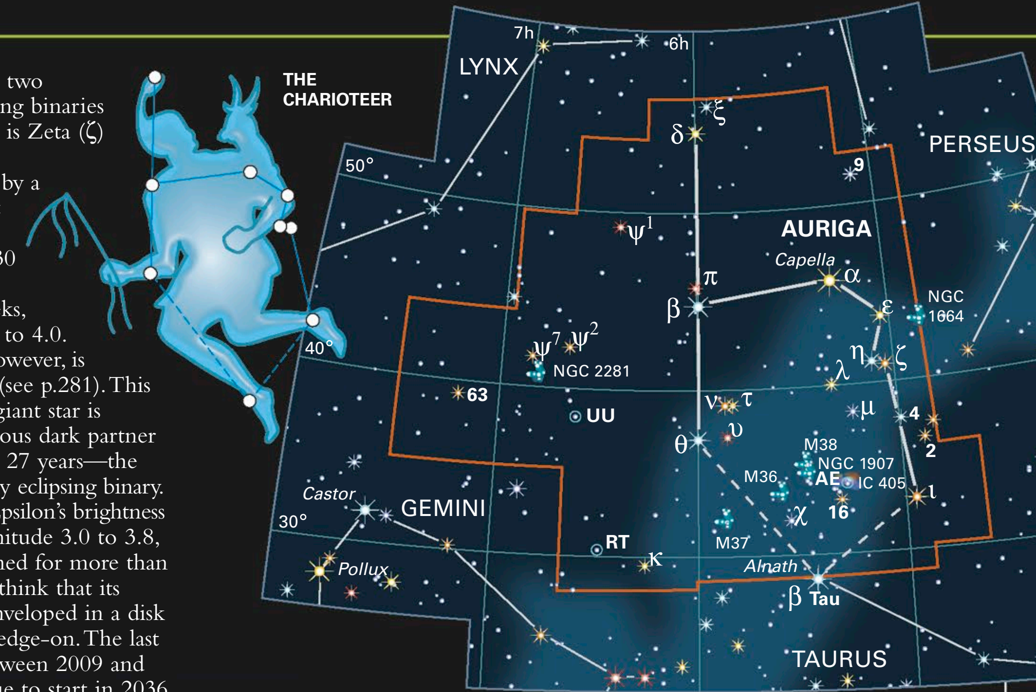
Auriga is easily identified in the northern sky by the presence of Capella (α), the most northerly first-magnitude star. Auriga lies in the Milky Way between Gemini and Perseus, to the north of Orion. The constellation represents a charioteer.

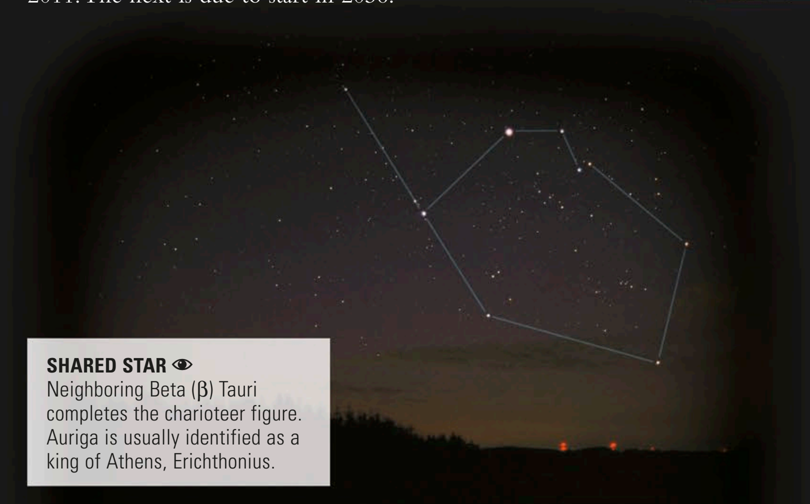
SPECIFIC FEATURES

Auriga's outstanding feature is a chain of three large and bright open star clusters. All three will just fit within the same field of view in wide-angle binoculars. Of the trio, M38's stars are the most scattered and, when viewed with a small telescope, seem to form chains. The middle cluster is M36, the smallest cluster but also the easiest to spot, while M37 is the largest and contains the most stars, but these are faint. All three clusters lie about 4,000 light-years away.

The star-forming nebula IC 405 is located nearby. Bright light from 6th-magnitude AE Aurigae near its center lights up the surrounding gases.

Auriga also contains two extraordinary eclipsing binaries of long period. One is Zeta (ζ) Aurigae, which is an orange giant orbited by a smaller blue star that eclipses it every 2.7 years. This causes a 30 percent decrease in brightness for 6 weeks, from magnitude 3.7 to 4.0. More remarkable, however, is Epsilon (ε) Aurigae (see p.281). This intensely luminous giant star is orbited by a mysterious dark partner that eclipses it every 27 years—the longest interval of any eclipsing binary. During the eclipse, Epsilon's brightness is halved, from magnitude 3.0 to 3.8, and it remains dimmed for more than a year. Astronomers think that its companion star is enveloped in a disk of dust seen almost edge-on. The last eclipse occurred between 2009 and 2011. The next is due to start in 2036.





SHARED STAR

Neighboring Beta (β) Tauri completes the charioteer figure. Auriga is usually identified as a king of Athens, Erichthonius.

THE FLAMING STAR NEBULA

AE Aurigae is a hot, massive star of magnitude 6 that lights up the surrounding cloud of gas and dust that is the Flaming Star Nebula, IC 405.

THE LYNX

Lynx


SIZE RANKING 28

BRIGHTEST STAR Alpha (α) 3.1

GENITIVE Lyncis

ABBREVIATION Lyn

HIGHEST IN SKY AT 10 P.M. February–March

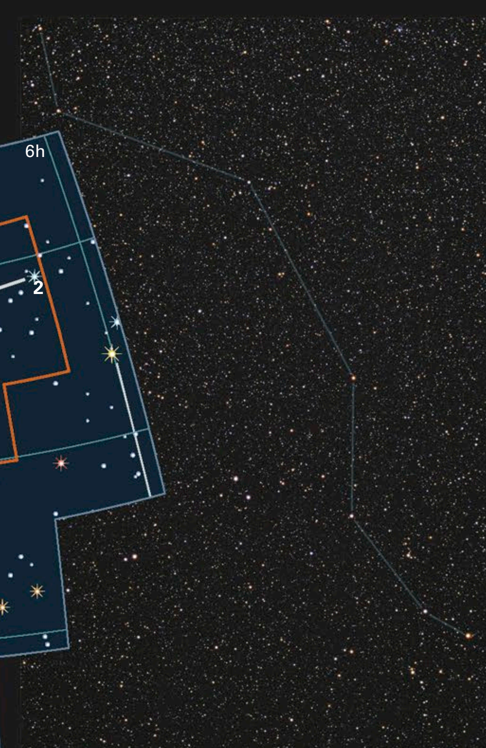
FULLY VISIBLE 90°N–28°S

Lynx is a fair-sized but faint constellation in the northern sky. It was introduced in the late 17th century by Johannes Hevelius (see p.384), who wanted to fill the gap between Ursa Major and Auriga. Hevelius is reputed to have named it Lynx because only the lynx-eyed would be able to see it—Hevelius himself had very sharp eyesight. The animal he drew on his star chart, however, looked little like a real lynx.

SPECIFIC FEATURES

Lynx contains many interesting double and multiple stars. For example, 12 Lyncis appears double with a small telescope, but with a telescope of 3 in (75 mm) or larger aperture the brighter star divides into two components of 5th and 6th magnitudes, which have an orbital period of about 900 years.

An easier triple to identify is 19 Lyncis. This consists of two stars of 6th and 7th magnitudes and a wider 8th-magnitude companion, all visible through a small telescope. A more challenging double star is 38 Lyncis, with components of 4th and 6th magnitudes. A telescope of 3-in (75-mm) aperture is required to separate the individual stars.



ELUSIVE FELINE

Lynx consists of nothing more than a few faint stars zigzagging between Ursa Major and Auriga. To spot it, sharp eyesight or binoculars are required.